Enhancing Elderly Safety: A Wearable Fall Detection System with IoT and Machine Learning

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***Abstract*—Falls represent a significant health concern, especially among the elderly and individuals with mobility impairments, often resulting in severe injuries or fatalities. Current fall detection solutions, such as webcam-based systems and manual alert devices, are often costly, raise privacy issues, and are prone to false alarms, highlighting a critical need for a more effective, reliable, and affordable solution. This project presents a wearable fall detection and emergency alert system integrating the ESP32 microcontroller, MPU6050 accelerometer and gyroscope, NEO-6M GPS module, and a Support Vector Machine (SVM) algorithm. The system aims to provide real-time fall detection and instant alerts to caregivers or emergency services, enhancing safety for vulnerable individuals, especially the elderly. The MPU6050 sensor monitors body movements, while SVM processes data to distinguish falls with 96% accuracy. The NEO-6M GPS module provides precise location tracking, and the ESP32 transmits alerts via Wi-Fi to the Blynk application, ensuring a timely emergency response. Comprehensive testing demonstrates high accuracy and reliability. This integration of machine learning, IoT, and advanced sensors offers an innovative and practical solution for fall detection in healthcare applications.**

Keywords— ***Fall Detection, Wearable Device, SVM, ESP32, IoT, GPS Tracking, Elderly Safety***

I. Introduction

Falls are a significant health concern, particularly for the elderly and individuals with disabilities, often leading to serious injuries or fatalities. In Malaysia, occupational injury cases increased by 13.8% in 2023, highlighting the need for improved fall prevention. Globally, the World Health Organization (WHO) reports 646,000 fatal falls annually, with adults over 65 years being the most affected. Falls not only impact individuals and families but also burden healthcare systems and society.

Existing fall detection methods, such as webcams and wearable emergency transmitters, have limitations, including high costs, privacy concerns, and frequent false alarms. Existing solutions also often struggle with real-time accuracy, suffer from frequent false alarms, or lack cost-effectiveness and ease of use, creating a significant research gap in practical wearable fall detection devices. This project proposes a cost-effective and reliable fall detection system using an accelerometer and gyroscope to monitor acceleration and body tilt. Mounting the device on the upper trunk minimizes false alarms and improves accuracy. Alerts are sent via SMS through Gmail to notify caregivers or healthcare centres for immediate assistance.

The novelty of this work lies in the integration of an ESP32 microcontroller, MPU6050 sensors, GPS module, and SVM-based machine learning, providing accurate, real-time fall detection and precise location tracking in a single wearable, cost-effective solution. This wearable system provides quick response times, reduces implementation costs, and enhances safety. Additionally, fall detection systems help reduce the fear of falling, which can otherwise lead to physical inactivity, depression, and lower quality of life. The proposed solution aims to mitigate fall-related risks and improve emergency response, ensuring better protection for at-risk individuals.

II. Related Works

The global population of older adults is rapidly increasing, with the number of people aged 60 and above projected to reach 2.1 billion by 2050. This demographic shift, largely driven by advancements in healthcare, poses significant public health challenges, especially regarding falls, which are the leading cause of injury-related hospitalizations in older adults. Falls can lead to severe injuries such as hip fractures, traumatic brain injuries, and upper limb injuries, significantly affecting healthcare costs and reducing mobility.

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Traditional fall detection methods, such as manual reporting, camera-based systems, and ambient sensor technology, each have limitations, including the need for active reporting, privacy concerns, and location constraints. Although webcam-based systems offer visual confirmation, they raise significant privacy concerns and have limited mobility. In contrast, wearable accelerometer-based systems provide better privacy but often experience higher false alarm rates due to their inability to effectively distinguish falls from other rapid movements. While wearable sensors, like single-axis accelerometers, are widely used, they often struggle with false alarms due to the inability to distinguish falls from other rapid movements [3], [5].

The advent of Internet of Things (IoT) technologies has improved fall detection systems by integrating wearable sensors, cloud computing, and wireless communication. The MPU6050, a sensor combining a 3-axis accelerometer and a 3-axis gyroscope, enhances fall detection by reducing false positives and offering real-time monitoring [1]. These IoT-enabled systems provide continuous fall detection, immediate alerts to caregivers and emergency services, and store historical data for trend analysis, ultimately improving fall risk assessments [4]. IoT-based wearable sensors offer higher accuracy, better mobility, and continuous monitoring, making them a preferred choice for elderly care and fall prevention.

IoT-based fall detection systems leverage sensors, connectivity, and data processing to enable real-time monitoring and emergency response in healthcare. The MPU6050 sensor, which integrates an accelerometer and gyroscope, detects motion and orientation changes, and transmits data via communication modules like Wi-Fi, Bluetooth, or LoRa to remote servers or cloud platforms. Data processing is handled by microcontrollers or cloud-based systems, which analyze movement patterns and trigger alerts when a fall is detected [9], [6].

These systems enhance traditional fall detection methods by providing automated alerts and remote access, allowing healthcare providers to monitor patients without physical presence [8]. Additionally, cloud-based storage enables long-term monitoring and predictive analytics to assess fall risks. By integrating IoT technology, fall detection systems offer greater efficiency, accuracy, and scalability, ultimately improving elderly care and reducing emergency response times [9]. Future advancements in AI-driven analytics and connectivity protocols are expected to further enhance the reliability and effectiveness of these systems. [10], [7]

Wearable devices have transformed health monitoring, fitness tracking, and motion analysis by integrating miniature sensors, wireless communication, and real-time data processing. These devices, such as smartwatches, fitness bands, and medical wearables, continuously track vital signs, movement patterns, and biomechanical parameters, making them crucial in healthcare, rehabilitation, and sports science [14]. The advancement of biomechanical analytics and sensor technology has enhanced the precision and reliability of wearable systems [11].

Wearable devices utilize various sensors, including Inertial Measurement Units (IMUs) like the MPU6050, which combine accelerometers and gyroscopes to track motion and angular velocity, providing accurate motion analysis [15]. Biosensors monitor physiological parameters such as heart rate, body temperature, and oxygen saturation, while Electromyography (EMG) sensors detect muscle activity, aiding in rehabilitation and performance monitoring [12].

In healthcare, wearables assist with continuous patient monitoring, fall detection, and rehabilitation tracking [19], while in sports, they help athletes improve techniques and reduce injury risks [13]. Wearable IMUs are also used in rehabilitation to monitor gait patterns and assess recovery progress [15]. The integration of wireless connectivity and data analytics has enhanced the accuracy and usability of these devices, making them indispensable in modern healthcare and fitness.

As technology advances, wearable sensors will continue to evolve, with improvements in miniaturization and battery life driving further innovation in health monitoring, fitness, and rehabilitation [14].

Machine learning (ML) has significantly enhanced fall detection accuracy by leveraging sensor data and advanced classification algorithms. Traditional threshold-based systems often struggle with false positives and misclassification of movements, prompting the integration of ML models for more precise pattern recognition and decision-making [20]. ML techniques, including deep learning, random forests, and convolutional neural networks (CNNs), have been widely applied to wearable-based fall detection, improving accuracy and reliability [18].

Supervised learning methods like random forests and support vector machines (SVMs) classify sensor data based on labelled fall and non-fall movements, while deep learning models such as Long Short-Term Memory (LSTM) and CNNs analyze sequential data for enhanced fall detection [21], [18]. Hybrid models that fuse sensor data and use multi-threshold judgment have also been explored to boost detection efficiency [19].

ML-based fall detection models have achieved accuracy rates exceeding 95%, particularly when feature selection and model training are optimized [17]. However, challenges such as computational complexity, real-time processing, and power consumption remain. Tiny Machine Learning (TinyML) is an emerging solution, enabling low-power ML models to run on edge devices, enhancing real-time detection while minimizing energy consumption [17]. SVMs have also been widely used in fall detection systems due to their ability to handle complex data, with studies showing that fusing sensor data with SVM algorithms can improve fall recognition rates and reduce false positives [19]. Overall, ML has greatly improved fall detection systems, but ongoing research is needed to enhance model generalization and optimize real-time performance in wearable devices. Unlike traditional wearable systems relying solely on accelerometer data, our approach leverages integrated gyroscope data and machine learning algorithms (SVM), significantly reducing false alarms and improving overall detection accuracy.

Emergency alert systems are crucial for public safety, disaster response, and healthcare monitoring, enabling rapid detection and communication during emergencies. These systems rely on sensor networks, wireless communication, and automation technologies to deliver real-time alerts to authorities, responders, and individuals at risk [25]. By improving response times, emergency alert systems help reduce injuries, fatalities, and damage in various scenarios, such as natural disasters, medical crises, and accidents [22].

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Modern systems combine hardware and software technologies, including microcontrollers, IoT devices, cloud computing, and wireless networks [27]. In disaster scenarios, seismic sensors and water level monitors predict hazards like earthquakes and floods, triggering alerts to residents and emergency services [26]. In road safety, motion sensors, GPS, and LoRa networks enhance response efficiency by notifying first responders of accidents [28]. In healthcare, emergency alerts assist patients with mobility challenges by detecting distress signals and notifying caregivers [24]. While emergency alert systems are widely used in disaster management, transportation safety, and healthcare, challenges like false alarms, network connectivity, and power consumption still need to be addressed. Advances in AI-driven analytics, low-power communication, and multi-sensor fusion are improving the reliability and scalability of these systems [22]. Ongoing research in predictive analytics, automation, and adaptive response strategies is expected to further enhance the effectiveness of real-time emergency communication systems.

III. Research Methodology

The flowchart in Fig. 1 outlines a real-time fall detection and emergency notification system. The process begins with the initialization of hardware components like sensors, microcontrollers, and communication modules. The system establishes a Wi-Fi connection through the Blynk app, allowing real-time data transmission and notifications. Motion data from the MPU6050 sensor is continuously collected to monitor movement, orientation, and velocity.

This data is processed by a pre-trained Support Vector Machine (SVM) algorithm, which distinguishes between normal activities and falls based on motion patterns. If no fall is detected, the system continues monitoring. If a fall occurs, the system collects GPS location data to pinpoint the user's whereabouts and uploads the alert and location to an IoT cloud platform for remote access.

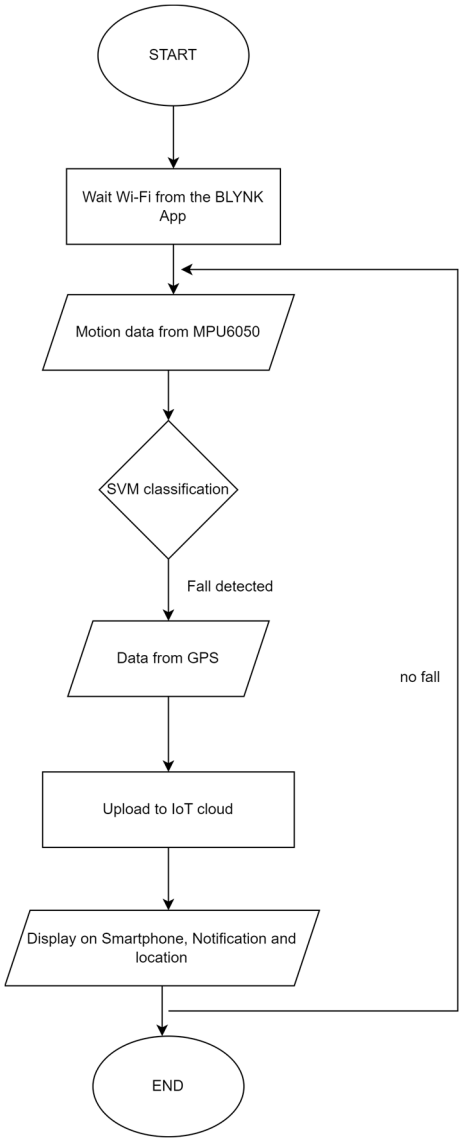


Fig. 1 Flowchart of the system

The alert, including the user's location, is sent to the user's smartphone via the Blynk app, notifying caregivers or emergency responders of immediate action. The system integrates motion sensors, machine learning, GPS tracking, and IoT technology to enhance safety, automate responses, and minimize delays, providing a reliable solution for vulnerable individuals like the elderly or those with mobility issues.

A. Hardware Configuration

Wrist-worn devices for fall detection are practical and comfortable, making them ideal for continuous monitoring, especially for the elderly or individuals with limited mobility. The wrist is a good location for fall detection because users often extend their arms to break a fall, allowing sensors to detect movements like sudden jerks or impacts. These devices use accelerometers and gyroscopes to measure changes in velocity and orientation, which help identify falls. The device incorporates key components to achieve real-time sensing, data processing, and emergency communication.

At the core of the device is the ESP32 microcontroller, a low-power, dual-core processor with integrated Wi-Fi and Bluetooth capabilities. It features multiple GPIO pins and supports various communication protocols, making it highly suitable for wearable IoT applications. Its energy-efficient design allows for extended operation on battery power, although the specific battery type and lifespan may vary depending on usage and update frequency. For this prototype, a 3.7V Li-ion rechargeable battery (1200mAh) was used, offering several hours of continuous monitoring and data transmission.

Two main sensors are interfaced with the ESP32:

- The MPU6050, a 6-axis motion tracking sensor that combines a 3-axis accelerometer and a 3-axis gyroscope, is responsible for capturing movement and orientation data. It communicates with the ESP32 via the I2C protocol, with its SCL pin connected to GPIO 22 and SDA to GPIO 21.
- The NEO-6M GPS module provides precise location tracking and is connected via UART communication, where its TX pin is connected to GPIO 16 (ESP32 RX), and RX is connected to GPIO 17 (ESP32 TX). This enables real-time location updates upon fall detection.

All components are powered through the ESP32's 3.3V output pin and grounded to the ESP32's GND pin. The

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compact and modular circuit ensures minimal wiring complexity, while also optimizing space and weight for wearable use. This hardware setup allows the ESP32 to process motion data, determine fall events using machine learning (detailed in the next section), and transmit alerts, along with GPS coordinates, via Wi-Fi using the Blynk IoT platform for real-time notification to caregivers or healthcare personnel. The combination of I2C and UART ensures stable, efficient data exchange without signal conflicts.

B. Fall Detection Algorithm and System Integration

Raw motion data from the MPU6050 sensor is pre-processed to ensure reliable input for the machine learning model. A low-pass filter removes high-frequency noise caused by vibrations or involuntary movements. Data normalization is applied through standard scaling to maintain consistency across all features.

From the filtered signals, three key time-domain features were extracted: peak accelerations, angular velocity thresholds, and duration of movement events. These features were selected based on their relevance to sudden movements and body orientation changes typical in fall scenarios. They served as the input feature vector for training the machine learning classifier to differentiate between fall and non-fall activities accurately.

A self-collected dataset was compiled from controlled experiments involving volunteers simulating various fall directions and non-fall daily activities. The dataset comprises 1700 labelled samples covering 13 activity types, including Fall Forward, Fall Backwards, Sitting, Standing, Walking, Stretching, and more. Each data sample includes six-axis motion data (3-axis accelerometer and 3-axis gyroscope readings). Feature vectors are constructed based on motion intensity and statistical descriptors extracted from each event.

A Support Vector Machine (SVM) classifier was developed using MATLAB's Classification Learner App. The dataset was split using 10-fold cross-validation to validate the model's performance and avoid overfitting. The model was evaluated based on precision, recall, F1-score, and overall accuracy. The final model achieved an accuracy of 96.53% with high reliability in detecting fall events.

Manual grid search was employed to optimize the SVM's hyperparameters for the best classification performance. The final model utilized a Radial Basis Function (RBF) kernel, which effectively captures nonlinear patterns in fall-related motion data. During tuning, different combinations of the box constraint (C) and kernel scale (gamma) were tested, and the optimal configuration was found to be C = 10 and gamma = 0.1. This combination provided a balance between maximizing the decision boundary margin and minimizing classification errors. The resulting model demonstrated high accuracy and computational efficiency, making it well-suited for real-time deployment on the resource-constrained ESP32 microcontroller.

After training, the SVM model was exported and embedded into the ESP32. The ESP32 continuously monitors sensor data, processes it using the trained SVM model, and triggers alerts when a fall is detected.

The system uses Blynk, an IoT platform that enables easy communication between hardware, cloud, and user interfaces. It offers a mobile app and dashboard for monitoring and controlling IoT devices. With its drag-and-drop interface, users can create custom dashboards with buttons, sliders, graphs, and notifications. In this project, Blynk connects the ESP32 microcontroller to send real-time alerts when a fall is detected, display GPS coordinates from the NEO-6M GPS module, and show sensor data from the MPU6050. Blynk's compatibility with various hardware and cloud integration makes it ideal for IoT applications.

IV. Results and Discussion

Fig. 2 shows the Blynk mobile interface used in the fall detection system, displaying both the real-time alert notification and the map-based GPS tracking. When a fall is detected, the system sends an immediate alert to the Blynk application, notifying the user or caregiver with relevant details such as the time and nature of the event. The GPS coordinates of the user's location are displayed on a map widget, allowing for quick identification and response. This integration enhances usability and ensures timely intervention, making it a crucial feature for improving user safety and caregiver responsiveness in real-world scenarios.

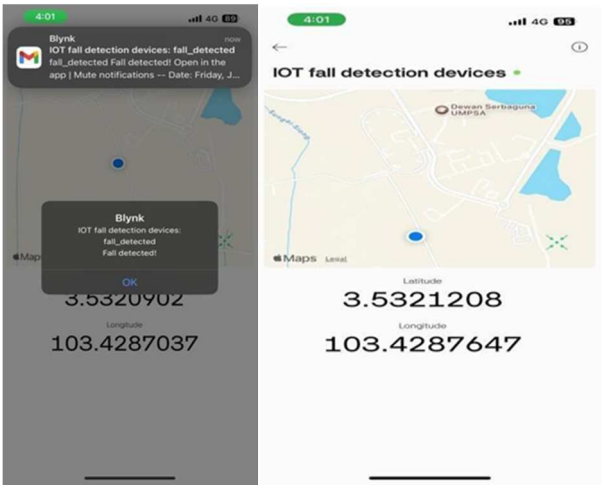


Fig. 2 Notification from Blynk

Fig. 3 shows the result of training SVM in a Matlab fall detection system, utilizing the Support Vector Machine (SVM) algorithm, which was evaluated for its effectiveness in distinguishing between falls and non-fall activities. The results, as indicated by the confusion matrix and performance metrics, demonstrate the system's strong performance:

```
SVM model training complete.
Evaluating model performance..
Confusion Matrix:
        1102          16
          29         148

Accuracy: 96.5251%
True Positives (TP): 148
True Negatives (TN): 1102
False Positives (FP): 16
False Negatives (FN): 29
Precision: 0.90244
Recall: 0.83616
F1-Score: 0.86804
```

Fig. 3 SVM training in Matlab

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Confusion Matrix:

- True Positives (TP): 148 falls correctly detected.
- True Negatives (TN): 1102 non-fall activities correctly classified.
- False Positives (FP): 16 non-falls misclassified as falls.
- False Negatives (FN): 29 falls not detected.

Key Metrics:

- Accuracy: 96.53%
  - The model accurately classified 96.53% of all instances, showcasing its overall reliability.
- Precision: 90.24%
  - Of all the predicted falls, 90.24% were actual falls, highlighting the system's low false alarm rate.
- Recall (Sensitivity): 83.62%
  - The system detected 83.62% of actual falls, indicating a strong ability to identify falls, though some were missed.
- F1-Score: 86.80%
  - The balance between precision and recall reflects a well-rounded performance

A. Data Analysis

As shown in Table 1, the performance analysis of the fall detection system revealed significant variations across the thirteen activity types tested. The system demonstrated exceptional accuracy in detecting fall events, with all four fall directions (Forward, Backward, Left, and Right) achieving 96% accuracy. Fall Forward and Fall Right activities showed the highest precision at 96.00%, with each recording 48 true positives and only 2 false positives. Activities related to falls consistently outperformed daily living activities, with precision rates ranging from 92.00% to 96.00% and recall rates between 79.31% and 82.76%. In contrast, non-fall activities such as Picking Up an Object and Stretching exhibited the lowest performance metrics, with precision and recall values of 69.09% and 65.52% respectively. Daily activities like Sitting Down, Lying, and Walking achieved moderate performance with accuracy rates around 82.00-84.00%. The overall pattern indicates that the SVM-based algorithm has been effectively optimized for its primary function of fall detection, while maintaining acceptable performance for classifying routine activities. This performance differential aligns with the system's intended application for emergency detection and response, where accurately identifying falls takes precedence over distinguishing between normal daily activities.

B. Receiver Operating Characteristic (ROC)

The performance of the fall detection model was primarily assessed using the Receiver Operating Characteristic (ROC) curve, as shown in Fig. 4. This curve plots the True Positive Rate (sensitivity) against the False Positive Rate, illustrating the trade-off between detecting actual fall events and avoiding false alarms. The model achieved an Area Under the Curve (AUC) of 0.910, indicating strong overall classification

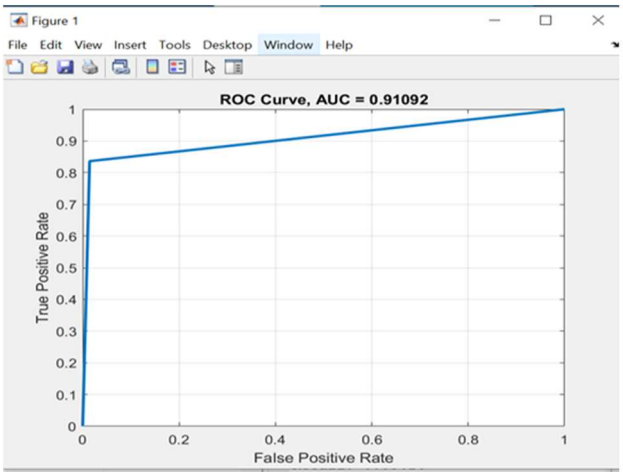


Fig. 4 The ROC curve

TABLE I. EVALUATION METRICS FOR ACTIVITY RECOGNITION USING MACHINE LEARNING

Activity type	True positive	False positive	True negative	False negative	precision	Recall	Accuracy
1. Fall Forward	48	2	190	10	96.00	82.76	**96.00**
2. Fall Backwards	47	3	189	11	94.00	81.03	**96.00**
3. Fall Left	46	4	188	12	92.00	79.31	**96.00**
4. Fall Right	48	2	190	10	96.00	82.76	**96.00**
5. Lying	40	15	177	18	72.73	68.97	82.80
6. Getting Up from chair	39	16	176	19	70.91	67.24	82.00
7. Lying Down	41	15	177	17	73.21	70.69	84.00
8. Picking Up an Object	38	17	175	20	69.09	65.52	82.00
9. Sitting Down	39	16	176	19	70.91	67.24	82.00
10. Sitting Still	40	15	177	18	72.73	68.97	82.80
11. Standing	41	15	177	17	73.21	70.69	84.00
12. Stretching	38	17	175	20	69.09	65.52	82.00
13. Walking	39	16	176	19	70.91	67.24	82.00

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performance. An AUC close to 1.0 reflects the model's high capability in distinguishing between fall and non-fall instances across various threshold settings. This result confirms that the proposed SVM-based fall detection system performs reliably, effectively balancing sensitivity and specificity, which is crucial in real-world safety applications.

V. Conclusion

This project successfully developed a wearable fall detection device using accelerometers and a gyroscope, along with a smartphone app for automatic alerts. The system achieved 96% accuracy in detecting falls through an SVM-based algorithm. The integration of the Blynk app provided real-time monitoring, alert notifications, and GPS

location tracking, ensuring caregivers or responders could quickly assist the user. Extensive testing and machine learning validation confirmed the system's effectiveness, demonstrating its potential to enhance safety and improve the quality of life for individuals at risk of falls.

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